

Roll No.

TPE-502

B. TECH. (PETROLEUM ENGINEERING) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

WELL LOGGING AND FORMATION EVALUATION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Write the importance of well logging in the E & P Industry. Write the history of the various logging tools and techniques adopted in the industry time to time. (CO1)

OR

- (b) Explain the principle of well logging techniques, what are the parameters measured by logging tools in the well ? Explain term vertical resolution and depth of investigation. (CO1)

2. (a) Explain the Physics of Gamma Ray (GR) tool with a neat diagram.

(CO2)

OR

- (b) Write formula for volume of shale estimation using GR Log. Calculate : (CO2)

$$V_{\text{Shale}} \text{ if } GR_{\text{Log}} = 40, GR_{\text{Shale}} = 100 \text{ \& } GR_{\text{Clean}} = 20.$$

P. T. O.

(2)

3. (a) Why is Caliper log is required ? Write the uses of caliper logs. Write the formula for calculation of mud cake thickness. (CO2)

OR

- (b) How the various types of lithology can be identify by caliper tool ? Explain with diagram. (CO2)

4. (a) Explain resistance and resistivity. Write formula for equivalent resistance in series and parallel combinations. (CO1, CO2)

OR

- (b) Write the principle of induction tool for resistivity measurement in the formation. Draw a neat diagram. (CO1, CO2)

5. (a) Write the principle of Density tool for Density measurement in the formation. Draw a neat diagram. (CO2)

OR

- (b) Drive the formula for estimation of porosity from density tool. Calculate porosity if $RHOB_Mat = 2.65$, $RHOB_Fluid = 1$, $RHOB_Bulk = 2.25$. (CO2)